

esade

PDS Final Project - Fast Food Companies: A Deep Performance Analysis

Date: November 20th, 2024

1 Dataset

In this final project, you will analyze the performance of the 10 big fast-food companies in the stock market over the years. To do so, you will have to form groups of 5 people and give answers to all the questions provided below.

Each dataset contains 10 csv files corresponding to a different fast food company's stock:

- **BRK-A**: Berkshire Hathaway Inc.
- **DNUT**: Krispy Kreme, Inc.
- **DPZ**: Domino's Pizza, Inc.
- **LKNCY**: Luckin Coffee Inc.
- **MCD**: McDonald's Corporation
- **PZZA**: Papa John's International, Inc.
- **QSR**: Restaurant Brands International, Inc.
- **SBUX**: Starbucks Corporation
- **WEN**: The Wendy's Corporation
- **YUM**: Yum! Brands, Inc.

On their side, columns represent the following:

- **Date**: Trading date
- **Open**: Opening price
- **High**: Highest price during session
- **Low**: Lowest price during session
- **Close**: Closing price
- **Adj Close**: Closing price adjusted for dividends and stock splits
- **Volume**: Number of shares traded during the session

REMEMBER: This project is not a project walkthrough, hence, first, you need to analyze carefully the datasets before going on with the exercises.

2 Exercises

You will have to give answers to the following 10 exercises (1 point each):

1: For each dataset:

- Print the number of rows and columns
- Display the column names and their data types

2: Extract all rows where the date is in the year 2023. Print the number of rows and visualize the **Close** price trend for this period.

3: For each company, find the day with the highest **Close** price and display it alongside the price.

4: Group the data by month and calculate the average **Close** price for each company. Plot these monthly averages for 3 companies and compare them. Justify the chart selection.

5: For each company:

- Compute the yearly average of the **Close** price
- Plot a comparison of yearly averages for all companies on a chart. Justify your chart selection

6: For each company create a plot showing the range of prices for each month and justify the chart selection.

7: Create a plot showing the relationship between trading volume and the **Close** price for a selected company. Add insights about the pattern, and, justify your chart selection.

8: For each company, identify the month with the highest total trading volume and display the results in a summary table, showing the month and total volume.

9: Merge the datasets for all companies into a single dataset, one for each year. Print the structure of the combined dataset, ensuring proper alignment and handling of missing values.

10: For each company:

- Calculate the spread between the **High** and **Low** prices for each day
- Compute the average spread for each company and visualize the result in a chart. Justify the chart selection
- Interpret which companies exhibit the largest spreads and why this might happen

3 What to Deliver

Each team should:

1. Create a new GitHub repository where all team members will have access.
2. Make sure all team members contribute to the project "committing" to the repository
3. Upload (also on GitHub) a the final presentation
4. **Deadline will be strictly followed. All projects should be uploaded on GitHub before December 14th, 23:59**